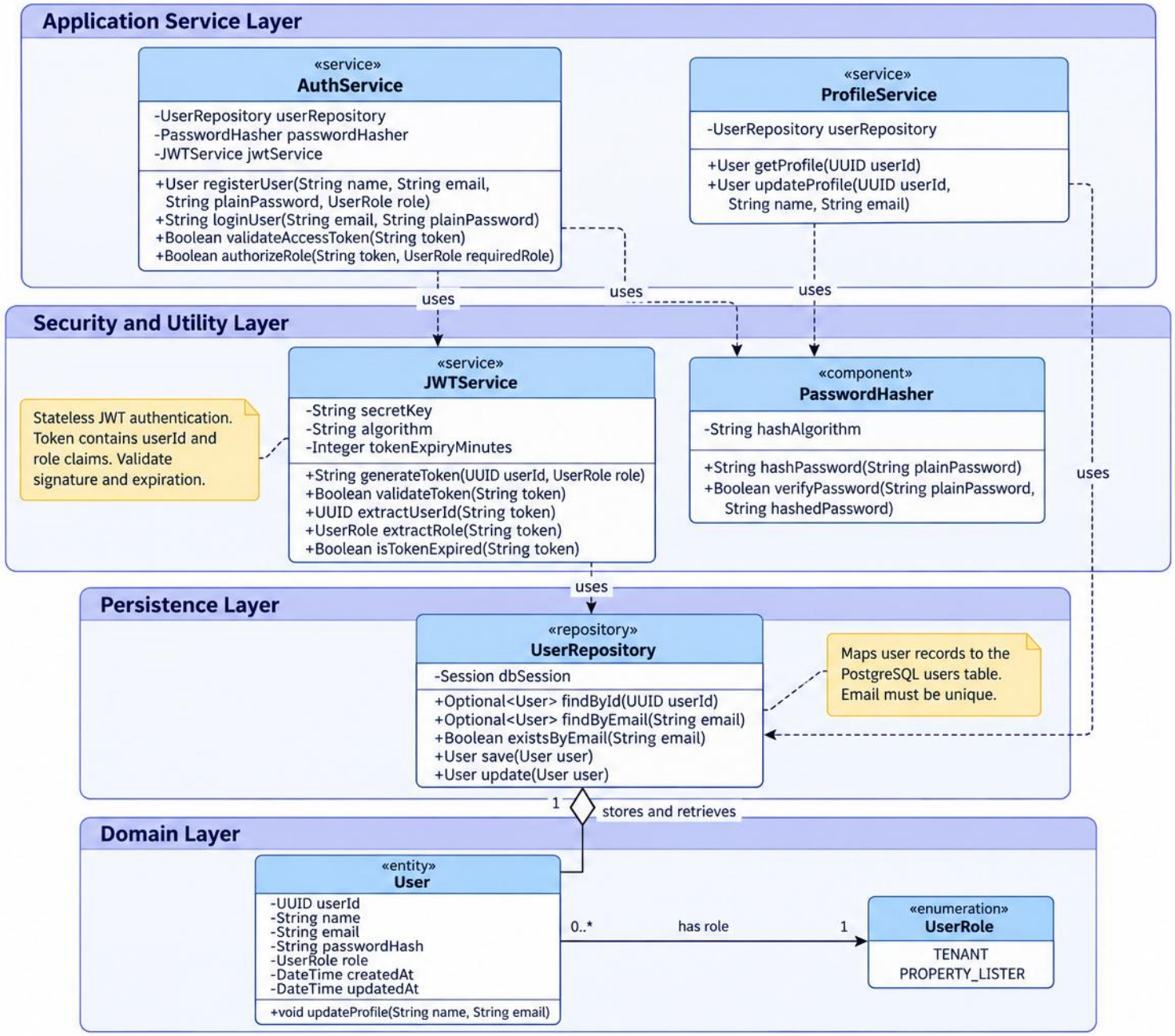

Application Service Layer
«service»
AuthService
-UserRepository userRepository
-PasswordHasher passwordHasher
-JWTService jwtService
+User registerUser(String name, String email, String plainPassword, UserRole role)
+String loginUser(String email, String plainPassword)
+Boolean validateAccessToken(String token)
+Boolean authorizeRole(String token, UserRole requiredRole)
«service»
ProfileService
-UserRepository userRepository
+User getProfile(UUID userId)
+User updateProfile(UUID userId, String name, String email)
uses
uses
uses
Security and Utility Layer
«service»
JWTService
-String secretKey
-String algorithm
-Integer tokenExpiryMinutes
+String generateToken(UUID userId, UserRole role)
+Boolean validateToken(String token)
+UUID extractUserId(String token)
+UserRole extractRole(String token)
+Boolean isTokenExpired(String token)
Stateless JWT authentication. Token contains userId and role claims. Validate signature and expiration.
«component»
PasswordHasher
-String hashAlgorithm
+String hashPassword(String plainPassword)
+Boolean verifyPassword(String plainPassword, String hashedPassword)
uses
uses
Persistence Layer
«repository»
UserRepository
-Session dbSession
+Optional<User> findById(UUID userId)
+Optional<User> findByEmail(String email)
+Boolean existsByEmail(String email)
+User save(User user)
+User update(User user)
Maps user records to the PostgreSQL users table. Email must be unique.
1
stores and retrieves
Domain Layer
«entity»
User
-UUID userId
-String name
-String email
-String passwordHash
-UserRole role
-DateTime createdAt
-DateTime updatedAt
+void updateProfile(String name, String email)
0..*
has role
1
«enumeration»
UserRole
TENANT
PROPERTY_LISTER